



# Modelling Financial Markets

## Chapter 4: Portfolio Optimisation



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## Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAPM: Capital Asset Pricing Model
- ▣ DSR: Deflated Sharpe Ratio
- ▣ ERC: Equal Risk Contribution portfolio
- ▣ FDR: False Discovery Rate
- ▣ FWER: Family-Wise Error Rate
- ▣ GLS: Generalised Least Squares
- ▣ GMV: Global Minimum-Variance portfolio
- ▣ GMV-LO: Global Minimum-Variance portfolio, Long-Only
- ▣ GMV-LW: Global Minimum-Variance portfolio with Ledoit-Wolf covariance
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HRP: Hierarchical Risk Parity
- ▣ HYG: iShares iBoxx High Yield Corporate Bond ETF
- ▣ IEF: iShares 7–10 Year Treasury Bond ETF
- ▣ LO: Long-Only, no short sales
- ▣ LQD: iShares iBoxx Investment Grade Corporate Bond ETF
- ▣ ML: Machine Learning
- ▣ MV: Mean–Variance tangency portfolio estimated from sample moments
- ▣ MV-LO: Mean–Variance tangency portfolio, Long-Only
- ▣ NYSE: New York Stock Exchange
- ▣ OLS: Ordinary Least Squares
- ▣ OMV: Österreichische Mineralölverwaltung (Austrian Mineral Oil Administration (OMV group))
- ▣ OOS: Out-Of-Sample

## Acronyms used in this material (2/2)

- POET: Principal Orthogonal compleMent Thresholding
- US: United States

## Chapter Route and Outcomes

### The question

- A **portfolio** is a weight vector  $w = (w_1, \dots, w_N)^\top$  with  $w^\top \mathbf{1} = 1$ ; optimisation chooses  $w$  to maximise a criterion under constraints
- The optimum is easy to write with known means and covariances, hard to estimate: the optimiser loads on the largest estimation errors
  - ▶ 16 assets:  $16 + 136 = 152$  parameters, estimated from 60 monthly observations of a 16-dimensional return vector
- Can any estimated rule beat equal weights out of sample, after costs?

### Route

- Part I: mean–variance theory, estimation risk, covariance shrinkage, Black–Litterman
- Part II: risk parity and HRP, out-of-sample tests, BVB portfolios, a case study, one pass of the AI-assisted research loop
- Technical detail sits in the [Appendix](#); textbook: [Franke, Härdle & Hafner \(2019\)](#), Part I

## Data, Assets and the 1/N Benchmark

| Universe    | Assets                                                                         | Returns           | Currency |
|-------------|--------------------------------------------------------------------------------|-------------------|----------|
| Sectors     | 9 US sector ETFs: XLB, XLE, XLF, XLI, XLK, XLP, XLU, XLV, XLY                  | 1999/01 – 2026/07 | USD      |
| Multi-asset | SPY, EFA, EEM (equities); TLT, IEF (Treasuries); LQD, HYG (credit); GLD (gold) | 2007/05 – 2026/07 | USD      |
| Combined    | 9 sectors + the 7 multi-asset ETFs other than SPY (16 ETFs)                    | 2007/05 – 2026/07 | USD      |
| Romania     | 8 BVB blue chips: TLV, SNP, BRD, TGN, SNG, SNN, EL, TEL; BET, BET-TR           | 2014/10 – 2026/08 | RON      |

- Monthly simple returns from month-end prices on common trading days; ETFs and stocks with dividends reinvested; BET is a price index, BET-TR reinvests dividends
- Excess returns over the one-month Treasury bill ([Kenneth French Data Library](#)); BVB:  $R_f = 0$
- Sectors: XLB materials, XLE energy, XLF financials, XLI industrials, XLK technology, XLP staples, XLU utilities, XLV health care, XLY discretionary
- Multi-asset: SPY (S&P 500), EFA (developed-market equities outside the US and Canada), EEM (emerging-market equities), TLT (US Treasuries, 20+ years), IEF (US Treasuries, 7–10 years), LQD (investment-grade corporate bonds), HYG (high-yield corporate bonds), GLD (gold)
- BVB: TLV Banca Transilvania, SNP OMV Petrom, BRD BRD–Groupe Société Générale, TGN Transgaz, SNG Romgaz, SNN Nuclearelectrica, EL Electrica, TEL Transelectrica
- 1/N: equal weights  $w_i = 1/N$ , rebalanced every month; no estimation at all, the benchmark of every comparison

## Notation

- Simple return  $R_{i,t} = P_{i,t}/P_{i,t-1} - 1$ ,  $P_{i,t}$  the month-end price of asset  $i$ ; excess return  $R_{i,t}^e = R_{i,t} - R_{f,t}$ ,  $R_{f,t}$  the risk-free rate
- $N$  assets;  $w$ : vector of portfolio weights;  $\mu$ : vector of expected (excess) returns;  $\Sigma$ : their  $N \times N$  covariance matrix;  $\mathbf{1}$ : vector of ones; portfolio mean  $w^\top \mu$ , variance  $w^\top \Sigma w$
- **Sharpe ratio** ([Sharpe \(1966\)](#)):  $SR_p = \mathbb{E}[R_p^e]/\sigma(R_p^e)$ , the mean excess return per unit of volatility  $\sigma$ ; monthly figures annualised with  $\times 12$  (mean) and  $\times \sqrt{12}$  (volatility, Sharpe)
- **Maximum drawdown**:  $\min_t (W_t / \max_{s \leq t} W_s - 1)$ , the largest fall of wealth  $W_t$  from a previous peak
- **Out of sample** (OOS): the weights held in month  $t$  use only data up to month  $t - 1$
- Rules of this chapter: MV (tangency from sample moments), GMV (global minimum variance), their long-only versions MV-LO and GMV-LO, GMV-LW (GMV with shrinkage), ERC, HRP and 1/N

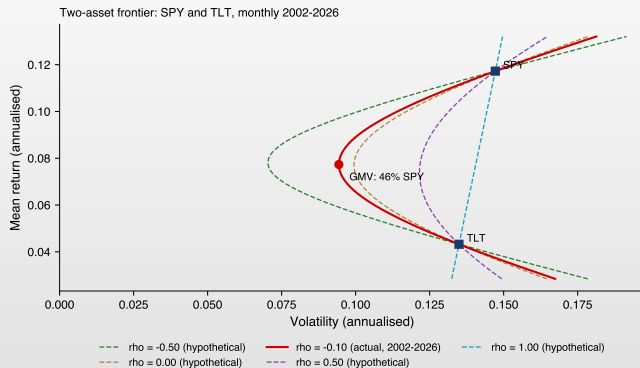
## The Markowitz Problem

- ▣ **Markowitz (1952)**: investors care about the mean and the variance of portfolio returns
- ▣ **Minimum-variance frontier**: minimum variance for every target mean  $m$ ; its upper branch,  $m \geq \mu_{\text{GMV}}$  (the mean of the GMV portfolio), is the **efficient frontier**

$$\min_w w^\top \Sigma w \quad \text{s.t.} \quad w^\top \mu = m, \quad w^\top \mathbf{1} = 1$$

- ▶ Objective: portfolio variance; constraints: the target mean  $m$  and full investment (weights sum to 1)
- ▣ Utility form, risk aversion  $\gamma > 0$ :  $\max_w w^\top \mu - \frac{\gamma}{2} w^\top \Sigma w$  s.t.  $w^\top \mathbf{1} = 1$ ; as  $\gamma$  varies it traces exactly the efficient branch
- ▣ Weights may be negative: a **short sale** (selling a borrowed asset)

## Two Assets: How Correlation Diversifies



- Two assets, correlation  $\rho$ :  

$$\sigma_p^2 = w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\rho\sigma_1\sigma_2$$
 $w$ : weight of asset 1,  $\sigma_1$ ,  $\sigma_2$ : volatilities,  $\sigma_p$ : portfolio volatility
- Minimised at  $w^* = \frac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$
- SPY and TLT, monthly 2002–2026:  
 $\rho = -0.10$ ,  $w^* = 46\%$  SPY, volatility 9.4% against 14.7% and 13.5%
- The lower  $\rho$ , the more the curve bends to the left; at  $\rho = 1$  there is no diversification
- Time-varying correlation: [Chapter 6](#)



## GMV and Tangency Portfolios for $N$ Assets

- GMV:  $\min_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w}$  s.t.  $\mathbf{w}^\top \mathbf{1} = 1$ ; Lagrangian  $\mathcal{L} = \mathbf{w}^\top \Sigma \mathbf{w} - \lambda(\mathbf{w}^\top \mathbf{1} - 1)$ ,  $\Sigma$  positive definite
  - ▶  $\lambda$ : Lagrange multiplier of the budget constraint; first-order condition  $2\Sigma \mathbf{w} = \lambda \mathbf{1}$ , so  $\mathbf{w} = \frac{\lambda}{2} \Sigma^{-1} \mathbf{1}$ ; the constraint gives  $\frac{\lambda}{2} = 1/A$ ,  $A = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}$
- Tangency: maximise  $\mathbf{w}^\top \boldsymbol{\mu} / \sqrt{\mathbf{w}^\top \Sigma \mathbf{w}}$ ,  $\boldsymbol{\mu}$  the expected **excess** returns; the condition  $\boldsymbol{\mu} \propto \Sigma \mathbf{w}$  gives  $\mathbf{w} \propto \Sigma^{-1} \boldsymbol{\mu}$  ( $\propto$ : proportional to);  $\theta$ : the maximum Sharpe ratio

|         | GMV                                                    | Tangency (MV)                                                                                                    |
|---------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Weights | $\mathbf{w}_{\text{GMV}} = \Sigma^{-1} \mathbf{1} / A$ | $\mathbf{w}_{\text{MV}} = \Sigma^{-1} \boldsymbol{\mu} / B$ , $B = \mathbf{1}^\top \Sigma^{-1} \boldsymbol{\mu}$ |
| Result  | variance $\sigma_{\text{GMV}}^2 = 1/A$                 | squared Sharpe $\theta^2 = \boldsymbol{\mu}^\top \Sigma^{-1} \boldsymbol{\mu}$                                   |
| Needs   | $\Sigma$ only                                          | $\boldsymbol{\mu}$ and $\Sigma$ ; Sharpe $\text{sign}(B) \theta$                                                 |

- The normalised  $\Sigma^{-1} \boldsymbol{\mu} / B$  is the maximum only if  $B > 0$ ;  $B < 0$  gives the minimum-Sharpe portfolio,  $B = 0$  no fully invested solution

## Separation and the Frontier in Closed Form

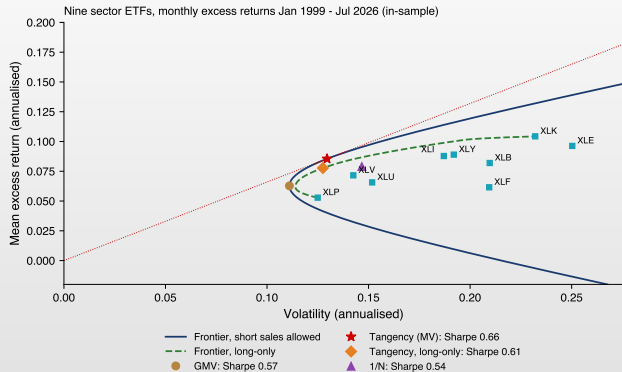


James Tobin, Nobel Prize in  
Economics 1981

Photo: unknown author (1962); public  
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- **Tobin (1958)**: add a riskless asset paying  $R_f$ ; every investor holds the **same** risky portfolio, the tangency portfolio, mixed with cash or borrowing
  - ▶ Risk aversion changes only the mix along the capital allocation line: **two-fund separation**, the bridge to the CAPM ([Chapter 3](#))
- **Merton (1972)**: with short sales, the frontier is a hyperbola;  $\Sigma$  positive definite,  $\mu$  not proportional to  $1$ 
  - ▶  $A = 1^\top \Sigma^{-1} 1$ ,  $B = 1^\top \Sigma^{-1} \mu$ ,  $C = \mu^\top \Sigma^{-1} \mu$ ,  $D = AC - B^2 > 0$
$$\sigma^2(m) = \frac{Am^2 - 2Bm + C}{D}$$
- $\sigma^2(m)$ : the smallest variance attainable with mean  $m$ , a parabola in  $m$ ; the GMV sits at  $m = B/A$ ,  $\sigma^2 = 1/A$ ; every frontier portfolio mixes any two distinct frontier portfolios, e.g. GMV and tangency (if  $B \neq 0$ )

## The Frontier of US Sectors



- Monthly excess returns, January 1999 – July 2026 ( $T = 331$ ); all parameters estimated on the full sample: an **in-sample** picture
- Tangency: Sharpe 0.66; GMV: 0.57 (volatility 11.1%); 1/N: 0.54
- Tangency weights:  $-0.37$  financials,  $-0.21$  materials,  $+0.37$  health care: short positions
- Long-only tangency: Sharpe 0.61; materials, financials and industrials get exactly zero, a **corner solution**
- The gain over 1/N is small even with perfect hindsight

## Constraints

- ▣ **Long-only (LO):**  $w_i \geq 0$ ; no closed form, solved by quadratic programming
- ▣ Position limits:  $w_i \leq 0.2$ ; sector or asset-class limits:  $\sum_{i \in G} w_i \leq c_G$ ,  $G$  a group of assets,  $c_G$  its cap
- ▣ Leverage limit:  $\sum_i |w_i| \leq L$ ;  $\sum_i |w_i|$ : gross exposure,  $L = 1$  means no short sales
- ▣ **Frost & Savarino (1988)**: constraining weights improved out-of-sample performance; why this works is a shrinkage result (next section)
- ▣ MV-LO and GMV-LO are the long-only versions of MV and GMV

## Means Are Hard to Estimate: Merton's Argument

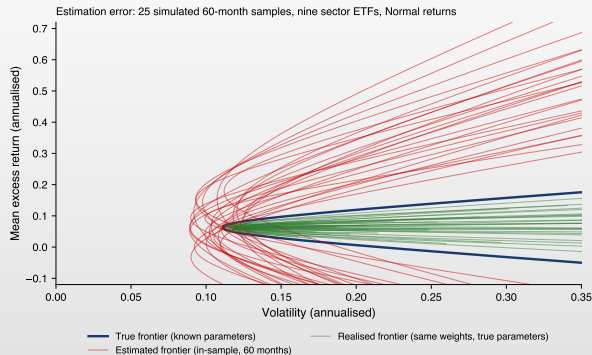


Robert C. Merton, Nobel Prize in  
Economics 1997

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- Merton (1980):  $Y$  years sampled  $h$  times a year, i.i.d. Normal returns (a diffusion) with annual mean  $\mu$  and variance  $\sigma^2$ 
  - ▶  $\text{Var}(\hat{\mu}) = \sigma^2/Y$ : independent of  $h$ ; only the **span** of the sample helps
  - ▶  $\text{Var}(\hat{\sigma}^2) \approx 2\sigma^4/(hY) \rightarrow 0$  as  $h \rightarrow \infty$  (Normal; in general  $(\kappa - 1)\sigma^4/(hY)$ ,  $\kappa$  the kurtosis): variances sharpen with frequency
- SPY,  $Y = 24$ ,  $\sigma = 14.7\%$ : standard error 3.0%, 95% interval [5.8%, 17.6%] for a mean of 11.7%
- A standard error of 1% needs  $(0.147/0.01)^2 \approx 217$  years
- Chopra & Ziemba (1993): errors in means cost about ten times more than errors in variances; rules without means (GMV, risk parity) are more robust

## Error Maximisation: Estimated versus Realised Frontiers



- **Michaud (1989):** optimisers are **estimation-error maximisers**; **Best & Grauer (1991):** a tiny change in one mean moves many weights
- True parameters = sector moments 1999–2026; draw 60 Normal months, optimise, evaluate with the true parameters
- Red: estimated frontiers, too optimistic; green: the same portfolios under the true parameters, all below the true frontier
- 2000 simulated tangency portfolios: median estimated maximum Sharpe 1.56, median true Sharpe 0.30, true optimum 0.66
- 96% have a true Sharpe below  $1/N$  (0.54), 7.5% below zero; in 11% of samples  $1^\top \hat{\Sigma}^{-1} \hat{\mu} < 0$

## The Plug-in Sharpe Ratio Is Biased

- i.i.d.  $N(\mu, \Sigma)$  returns, maximum-likelihood  $\hat{\mu}$ ,  $\hat{\Sigma}$ ,  $\hat{\theta}^2 = \hat{\mu}^\top \hat{\Sigma}^{-1} \hat{\mu}$ , the in-sample squared maximum Sharpe ratio;  $T$  months,  $N$  assets (Kan & Zhou (2007); Wishart moments in the [Appendix](#))

$$\mathbb{E}[\hat{\theta}^2] = \frac{T}{T - N - 2} \left( \theta^2 + \frac{N}{T} \right) = \frac{T\theta^2 + N}{T - N - 2}, \quad T > N + 2$$

- Sectors:  $N = 9$ ,  $T = 60$ , true  $\theta = 0.66$  a year:  $\mathbb{E}[\hat{\theta}^2]$  corresponds to a Sharpe of 1.65 a year, the analytical counterpart of the simulated median 1.56

- Unbiased estimator of  $\theta^2$ :  $\hat{\theta}_u^2 = [(T - N - 2)\hat{\theta}^2 - N] / T$

► It can be negative; truncating at zero gives a nonnegative but upward-biased estimator, and  $\sqrt{\hat{\theta}_u^2}$  is not unbiased for  $\theta$

- Sectors, full sample ( $T = 331$ ): plug-in  $\hat{\theta} = 0.66$ , bias-corrected  $\sqrt{\hat{\theta}_u^2} = 0.31$  a year: more than half of the in-sample Sharpe is noise, even with 27 years

- The tangency weights are OLS coefficients of a regression of 1 on the returns (Britten-Jones (1999); [Appendix](#)), Seminar A2 and B9)

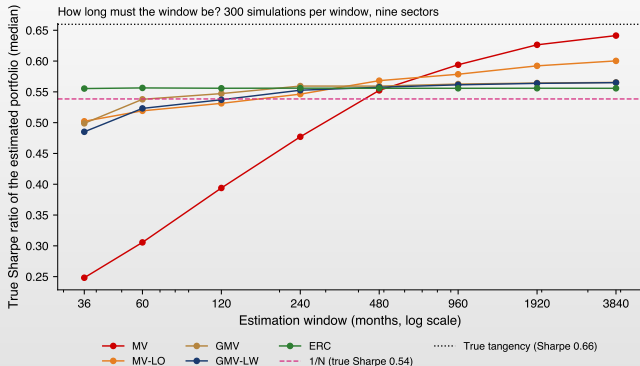
## Unstable Weights



- Nine sector ETFs; each month the weights use only the previous 60 months; top panel clipped at  $\pm 6$
- MV: median gross exposure  $\sum_i |w_i| = 4.9$ , above 3 in 90% of the months; largest single weight 26 in absolute value
- Explosions in 2004–2006 and 2009: the denominator  $1^\top \hat{\Sigma}^{-1} \hat{\mu}$  is close to zero, so the tangency weights have very heavy tails
  - ▶ Negative in 12 of 271 windows: MV then holds the minimum-Sharpe portfolio
- GMV-LW (bottom): gross exposure 1.5, largest weight 0.94
- Unstable weights mean high turnover, hence high trading costs



## How Long Must the Window Be to Beat 1/N?



- DeMiguel, Garlappi & Uppal (2009): 14 models on 7 data sets; none beats 1/N consistently in Sharpe ratio, certainty-equivalent return or turnover; their calibration needs about 3000 months for 25 assets
- Here: 300 simulated samples per window, population parameters = sector moments
- MV beats 1/N only with 480 months (40 years): median true Sharpe 0.55 vs 0.54; with 60 months 0.31
- MV-LO, GMV, GMV-LW are close to 1/N from 60 months on; ERC is almost insensitive, 0.56
- In this calibration 1/N is nearly efficient (true tangency 0.66): simulation evidence, not an empirical test

 MFM\_ch4\_window\_length

## Why the Sample Covariance Matrix Fails

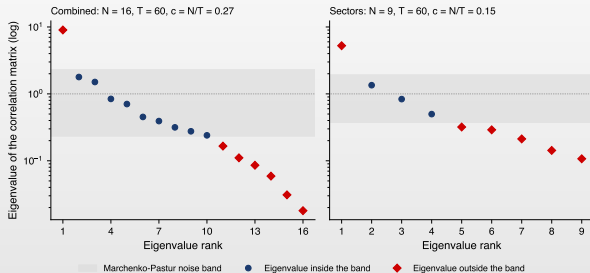
- ▣ Sample eigenvalues are too dispersed: the largest too large, the smallest too small
- ▣ **Condition number**: largest over smallest eigenvalue; a large value means  $\Sigma^{-1}$  amplifies noise
- ▣ Optimisers invert  $\Sigma$ , so they load on the directions with the smallest estimated variance
- ▣ The demeaned sample matrix has rank at most  $T - 1$ : singular, not invertible, whenever  $T \leq N$
- ▣ **Marchenko–Pastur** benchmark ([Marčenko & Pastur \(1967\)](#)): i.i.d. returns with  $\Sigma = \sigma^2 I$ ; as  $N, T \rightarrow \infty$  with  $N/T \rightarrow c \in (0, 1)$ , the sample eigenvalues fill

$$[\lambda_-, \lambda_+] = [\sigma^2(1 - \sqrt{c})^2, \sigma^2(1 + \sqrt{c})^2]$$

- ▶  $I$ : identity matrix;  $\lambda_-$ ,  $\lambda_+$ : the smallest and largest eigenvalue that pure noise produces;  $c = 16/60 = 0.27$ : band  $[0.23, 2.30]$  for a correlation matrix ( $\sigma^2 = 1$ ), a ten-fold spread from noise alone
- ▶ [Laloux et al. \(1999\)](#): eigenvalues outside the band are candidates for structure; those inside cannot be told from noise by this comparison alone

## Sample Eigenvalues versus the Noise Band

Last 60-month window, Aug 2021 - Jul 2026



- Correlation matrices of monthly excess returns, last 60-month window (August 2021 – July 2026); shaded band:  $[\lambda_-, \lambda_+]$  for  $\sigma^2 = 1$
- Only the market mode lies above  $\lambda_+$ : 9.0 (56% of the trace) for 16 ETFs, 5.2 for nine sectors ( $\lambda_+ = 1.92$ ); with the noise variance rescaled to  $\sigma^2 = 1 - \lambda_1/N$  (Laloux et al. (1999)), three eigenvalues exceed the band in each universe
- Six of 16 eigenvalues lie **below** the band: near-riskless combinations such as TLT against IEF; GMV loads exactly on them, so if they are underestimated, so is its risk

## The Shrinkage Estimator

- Shrink the sample matrix  $S$  towards a structured target  $F$

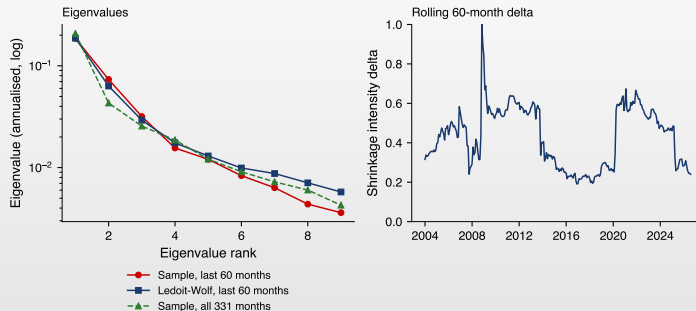
$$\hat{\Sigma}_{\text{LW}} = \delta F + (1 - \delta) S, \quad 0 \leq \delta \leq 1$$

- Targets

- ▶ Single-index (market) model: Ledoit & Wolf (2003)
- ▶ Scaled identity: Ledoit & Wolf (2004a)
- ▶ Constant correlation,  $f_{ii} = s_{ii}$  and  $f_{ij} = \bar{r} \sqrt{s_{ii} s_{jj}}$  for  $i \neq j$ ,  $\bar{r}$  the average sample correlation: Ledoit & Wolf (2004b), used here

- $\delta$  trades the variance of  $S$  against the bias of  $F$ : noisy data or a good target, large  $\delta$ ; long samples,  $\delta \rightarrow 0$  (the optimal  $\delta^*$  in the [Appendix](#))

## Shrinkage on US Sectors



- Nine sector ETFs, monthly excess returns; last window August 2021 – July 2026:  $\delta = 0.24$ ; the smallest eigenvalues are pulled up, the largest almost unchanged; condition number  $51.6 \rightarrow 32.5$
- Rolling  $\delta$ : mean 0.43, range 0.19–1.00;  $\delta = 1$  in the early windows, where 60 months are too few to beat the target
- GMV-LW: the GMV portfolio computed with  $\hat{\Sigma}_{LW}$

## Constraints Are Shrinkage Too

- ▣ Jagannathan & Ma (2003): the long-only GMV equals the unconstrained GMV of a modified covariance matrix

$$\tilde{S} = S - (\lambda \mathbf{1}^\top + \mathbf{1} \lambda^\top), \quad \mathcal{L} = \frac{1}{2} \mathbf{w}^\top S \mathbf{w} - \lambda_0 (\mathbf{w}^\top \mathbf{1} - 1) - \lambda^\top \mathbf{w}$$

- ▶  $\lambda \geq 0$ : the Lagrange multipliers of the constraints  $w_i \geq 0$  in  $\mathcal{L}$ ;  $\lambda_i > 0$  only where the constraint binds
- ▶ The first-order condition  $S \mathbf{w} = \lambda_0 \mathbf{1} + \lambda$  and  $\lambda^\top \mathbf{w} = 0$  give  $\tilde{S} \mathbf{w} = \lambda_0 \mathbf{1}$ : the GMV condition for  $\tilde{S}$
- ▣  $\tilde{s}_{ij} = s_{ij} - \lambda_i - \lambda_j$ : a binding constraint on asset  $i$  lowers its covariances, which shrinks the large estimates
- ▣ Imposing the “wrong” constraint (no short sales) can reduce out-of-sample risk; the effect is largest when covariances come from few observations

## Nonlinear Shrinkage

- Linear shrinkage mixes two matrices with one  $\delta$ ; **nonlinear** shrinkage keeps the sample eigenvectors and corrects each eigenvalue separately
  - ▶ Ledoit & Wolf (2017): tuned for portfolio selection; Ledoit & Wolf (2020): an analytical formula, no optimisation
- Out of sample, 60-month window, GMV volatility
  - ▶ Sectors: sample 11.90%, linear (constant correlation) 11.80%, nonlinear 11.62%
  - ▶ 16 ETFs: sample 4.06%, linear 4.38%, nonlinear 4.31%: better than the linear target, not than the raw matrix
- A constant correlation fits neither stock–stock (positive) nor stock–bond (negative) pairs: the target matters (Seminar 4, B4)
- Rule: shrink, constrain, or both, but never invert a raw sample matrix of many assets

## From Covariance Shrinkage to Mean Shrinkage



Fischer Black (1938–1995)

Photo: Dalmatine at English

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- Shrinkage fixes  $\hat{\Sigma}$ ; Black & Litterman (1992), developed at Goldman Sachs, shrinks the means towards those implied by a **benchmark** portfolio  $w_b$
- Reverse optimisation: a mean–variance investor with risk aversion  $\delta_{\mathbf{RA}}$  and access to cash holds  $w^* = (\delta_{\mathbf{RA}} \Sigma)^{-1} \mu$  in risky assets; the returns that make  $w_b$  optimal are

$$\pi = \delta_{\mathbf{RA}} \Sigma w_b$$

- $\pi$ : **implied** (equilibrium) excess returns; with market weights, the CAPM returns
- Here: no market capitalisations for the sectors, so  $w_b = 1/N$ ,  $\delta_{\mathbf{RA}} = 2.5$
- Then add the investor's **views**, weighted by their confidence: the weights move away from the benchmark only where views exist

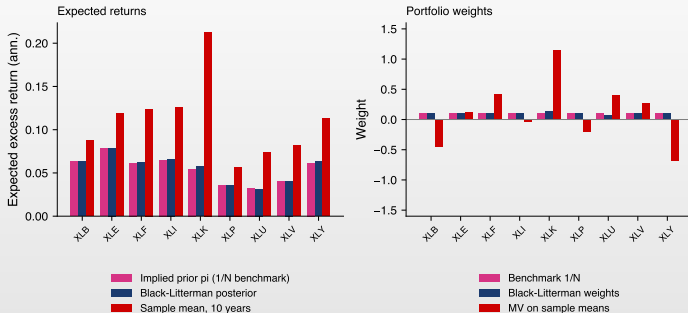


## Views and the Posterior

- $K$  views:  $P\mu = q + \varepsilon$ ,  $\varepsilon \sim N(0, \Omega)$ ; prior  $\mu \sim N(\pi, \tau\Sigma)$ 
  - $P$ :  $K \times N$  matrix, row  $k$  = the portfolio of view  $k$ ;  $q$ : the view returns;  $\Omega$ : their uncertainty;  $\tau$ : scales the uncertainty of the prior  $\pi$
- Posterior mean: a precision-weighted average; weights  $w_{BL} = (\delta_{RA}\Sigma)^{-1}\mu_{BL}$ 

$$\mu_{BL} = [(\tau\Sigma)^{-1} + P^\top \Omega^{-1}P]^{-1}[(\tau\Sigma)^{-1}\pi + P^\top \Omega^{-1}q]$$
- Limits:  $\Omega \rightarrow 0$  gives  $P\mu_{BL} = q$  exactly;  $\Omega \rightarrow \infty$  gives  $\pi$ ; without views, the benchmark itself
- He & Litterman (2002):  $\tau = 0.05$ ,  $\Omega = \text{diag}(P \tau \Sigma P^\top)$ ; with  $\Sigma$  in the weights, only the assets in the views move
- One asset,  $\pi = 5\%$ ,  $\tau\sigma^2 = 0.00128$ , view 8% with variance 0.0004:  $\mu_{BL} = 7.29\%$ , view weight 0.76
- The same estimator is a GLS (mixed) regression (Theil & Goldberger (1961); [Appendix](#), Seminar A5)

## Black–Litterman on US Sectors



- $\Sigma$  from monthly excess returns, August 2016 – July 2026; one view: XLK minus XLU = 3% a year; implied 2.2%, posterior 2.6% (the average, since  $\Omega$  equals the prior variance of the view)
- Weights: XLK from 11.1% to 14.4%, XLU from 11.1% to 7.8%, all others unchanged; the view is relative ( $P1 = 0$ ), so the tilt  $\propto P^\top$  sums to zero and the portfolio stays fully invested
- Sample means instead: XLK 21.3% a year; MV weights +115% XLK, -68% XLY, -46% XLB

## Risk Contributions

- Volatility is homogeneous of degree one in the weights (Euler)

$$\sigma_p = \sum_{i=1}^N w_i \frac{\partial \sigma_p}{\partial w_i} = \sum_{i=1}^N \frac{w_i (\Sigma w)_i}{\sigma_p}$$

- Risk contribution** of asset  $i$ :  $RC_i = w_i (\Sigma w)_i / \sigma_p$ ; percentage share  $RC_i / \sigma_p = w_i (\Sigma w)_i / \sigma_p^2$
- Worked example, 50/50 in SPY and IEF (monthly 2002–2026, annualised):  $\sigma_1^2 = 0.02166$ ,  $\sigma_2^2 = 0.00441$ ,  $\sigma_{12} = -0.00098$ 
  - $(\Sigma w)_1 = 0.5(0.02166) + 0.5(-0.00098) = 0.01034$ ,  $(\Sigma w)_2 = 0.5(-0.00098) + 0.5(0.00441) = 0.00172$
  - $\sigma_p^2 = 0.5(0.01034) + 0.5(0.00172) = 0.00603$ ,  $\sigma_p = 7.76\%$ ; shares: SPY 85.8%, IEF 14.2%
- Equal weights are not equal risks: volatile assets dominate; risk budgeting chooses  $w$  so that the shares equal preset budgets  $b_i$

## Equal Risk Contribution

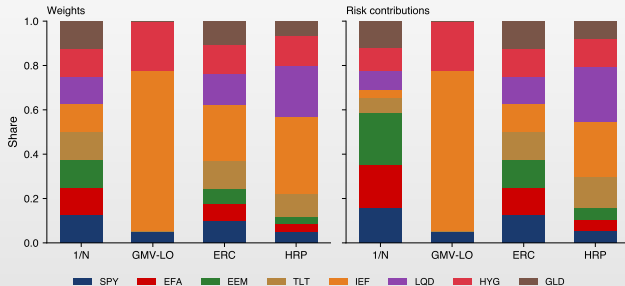
- ERC (Maillard, Roncalli & Teïletche (2010)):  $RC_i = RC_j$  for all  $i, j$ ; computed from a convex problem, then normalised

$$\min_{y>0} \frac{1}{2} y^\top \Sigma y - \frac{1}{N} \sum_i \ln y_i, \quad w = y / 1^\top y$$

- $y$ : unnormalised positive weights; the log term keeps every  $y_i > 0$  and its first-order condition gives  $y_i(\Sigma y)_i = 1/N$  for all  $i$
- For  $\Sigma$  positive definite a solution exists and is unique (strictly convex objective, infinite at the boundary)
- Volatility ordering:  $\sigma_{\text{GMV}} \leq \sigma_{\text{ERC}} \leq \sigma_{1/N}$
- Two assets: ERC is **inverse volatility**,  $w_1 = \sigma_1^{-1} / (\sigma_1^{-1} + \sigma_2^{-1})$ , for any  $\rho \in (-1, 1)$ 
  - SPY (14.7%) and IEF (6.6%): ERC holds 31% SPY, volatility 6.1%, each asset 50% of the variance
- Needs no expected returns, only  $\Sigma$ ; ERC is the tangency portfolio if all assets have the same Sharpe ratio and equal correlations

## Weights versus Risk Contributions

Eight multi-asset ETFs, monthly May 2007 - Jul 2026

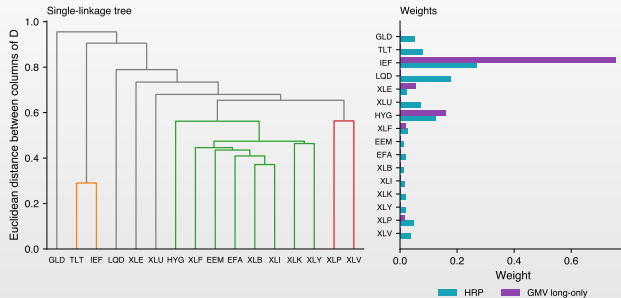


- Eight multi-asset ETFs, full-sample covariance matrix, May 2007 – July 2026 ( $T = 231$ )
- 1/N: equities and high yield receive half of the weight but 69% of the risk; ERC: 12.5% of the risk from every asset, IEF gets the largest weight (25%); long-only GMV: 73% IEF, 22% HYG, 5% SPY, five assets at zero
- Volatility: 1/N 9.4%, ERC 7.9%, HRP 7.1%, long-only GMV 5.8%: ERC lies between 1/N and GMV, as the theory predicts

## Hierarchical Risk Parity: From a Tree to Weights

- HRP (López de Prado (2016)), three steps, no matrix inversion (works even when  $\Sigma$  is singular), all weights positive
  1. **Tree**: correlation distance  $d_{ij} = \sqrt{(1 - \rho_{ij})/2} \in [0, 1]$ ; then the Euclidean distance between the **columns** of  $D = (d_{ij})$ ,  $\tilde{d}_{ij} = \sqrt{\sum_k (d_{ki} - d_{kj})^2}$ ; single linkage on  $\tilde{d}_{ij}$  (repeatedly merge the two closest clusters) gives a **dendrogram**
  2. **Quasi-diagonalisation**: order the assets as the leaves of the tree, so that similar ones are adjacent
  3. **Recursive bisection**: split the **ordered list** in two halves; give each half weight in inverse proportion to its variance  $\tilde{w}^\top \Sigma \tilde{w}$ , with  $\tilde{w}$  the inverse-variance weights inside the half; repeat inside each half
- Mantegna (1999): the same distance and the **minimum spanning tree** reveal the hierarchy of a stock market; textbook: Härdle & Simar (2019), cluster analysis
- A worked four-asset example: Seminar 4, B3

## HRP on 16 ETFs



- Nine sectors and seven multi-asset ETFs, monthly, May 2007 – July 2026; closest pair TLT and IEF ( $\tilde{d} = 0.29$ ), gold joins last (0.96)
- Equity cluster: cyclical sectors, EFA, EEM and high-yield credit; defensive sectors (utilities, staples, health care) and energy join it late
- HRP: all 16 weights positive, largest IEF 26.7%, LQD 17.6%, HYG 12.6%; long-only GMV: 75.6% in IEF and only five assets above zero

## Backtest Design

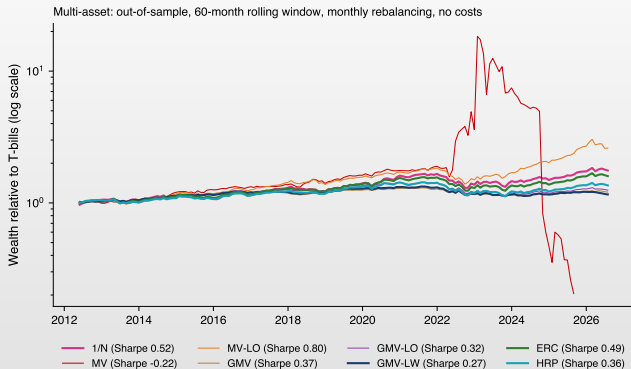
- ▣ Rolling window of 60 months, monthly rebalancing; the weights of month  $t$  use months  $t - 60, \dots, t - 1$  only
- ▣ Eight rules: 1/N, MV, MV-LO, GMV, GMV-LO, GMV-LW, ERC, HRP
- ▣ Three universes
  - ▶ Sectors: OOS January 2004 – July 2026, 271 months (first window January 1999 – December 2003)
  - ▶ Multi-asset and combined: OOS May 2012 – July 2026, 171 months
- ▣ Annualised mean, volatility and Sharpe ratio of excess returns; maximum drawdown of total-return wealth
- ▣ Every Sharpe ratio difference is reported with a test (next slides), and every test run enters the multiple-testing correction



## Turnover and Costs: One Worked Month

- Between rebalancings weights drift with returns:  $w_{i,t}^+ = w_{i,t}(1 + R_{i,t}) / \sum_j w_{j,t}(1 + R_{j,t})$
- **Turnover:** traded fraction of wealth,  $TO_{t+1} = \sum_i |w_{i,t+1} - w_{i,t}^+|$
- Worked month, sectors, 1/N: in January 2004 XLB returned  $-4.4\%$  and XLK  $+4.3\%$ , so the weights of 11.11% drifted to 10.50% and 11.45%
  - ▶ Rebalancing back to 1/N in February 2004 trades  $TO = 0.019$ : 1/N also trades, selling winners and buying losers
  - ▶ MV-LO, re-estimated on the new window, trades 0.748 in the same month
- Net return with cost  $c$  per unit traded:  $R_{p,t}^{\text{net}} = R_{p,t} - c TO_t$ ; 1 basis point (bp) = 0.01%, so 20 bp means  $c = 0.002$ 
  - ▶ At 20 bp the February trade costs MV-LO  $0.002 \times 0.748 = 0.15\%$  of wealth, 1/N 0.004%

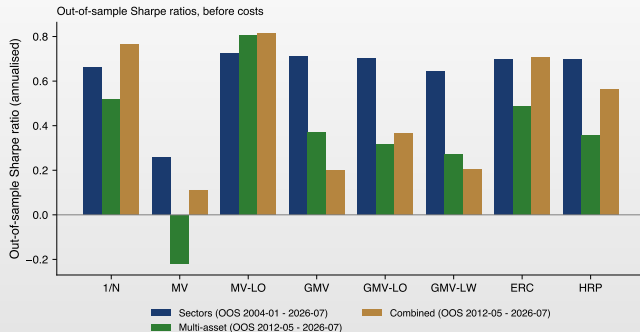
## Out-of-Sample Growth: Multi-Asset



- Eight multi-asset ETFs; wealth relative to a Treasury-bill account, before costs
- MV collapses: volatility 518% a year, turnover 44 times wealth per month; a monthly loss above 100% ends its wealth path
- MV-LO has the highest Sharpe (0.80); 1/N 0.52; ERC 0.49
- GMV rules: lowest volatility (3.8%–5.4%), low Sharpe (0.27–0.37); they load on Treasuries, whose excess returns were slightly negative after May 2012 (TLT  $-0.5\%$ , IEF  $-0.2\%$  a year)
- Is MV-LO really better, or lucky? A test is needed

 MFM\_ch4\_oos\_backtest

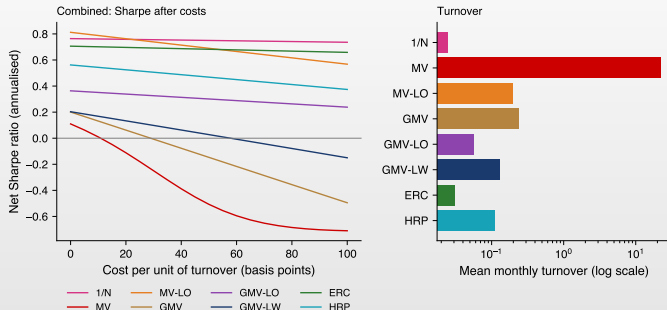
## Sharpe Ratios Across Universes



- 60-month rolling window, monthly rebalancing, annualised, before costs; complete tables in the [Appendix](#)
- Sectors: every rule except MV lands within 0.07 of 1/N (0.66); MV 0.26 with drawdown -100%
- Combined: 1/N 0.76, MV-LO 0.81, ERC 0.71; the GMV rules 0.20–0.36
- MV: a monthly loss above 100% wipes out the capital; its mean and Sharpe describe a hypothetical return series, not investable wealth
- Turnover per month, sectors: 1/N 0.02, ERC 0.03, GMV-LW 0.14, MV-LO 0.23, MV 3.86

 [MFM\\_ch4\\_oos\\_backtest](#)

## Sharpe Ratios after Costs



- Combined universe (16 ETFs); cost  $c$  from 0 to 100 bp per unit of turnover; turnover from 0.02 (1/N) to 22 (MV) per month
- Net Sharpe turns negative at about 11 bp for MV, 29 bp for GMV, 58 bp for GMV-LW; MV-LO overtakes 1/N only for costs below about 22 bp (Seminar 4, B8)
- 1/N and ERC barely move: low turnover is a hidden advantage; report net performance for realistic costs, not only gross

## Inference on One Sharpe Ratio: Lo (2002)

- ▣ Lo (2002), i.i.d. Normal returns:  $\sqrt{T}(\widehat{SR} - SR) \rightarrow N(0, 1 + SR^2/2)$ , with  $SR$  monthly,  $T$  the number of months: the standard error is  $\sqrt{(1 + SR^2/2)/T}$
- ▣ 1/N on sectors,  $T = 271$ : monthly  $\widehat{SR} = 0.191$ , standard error 0.061
  - ▶ Annualised  $0.66 \pm 1.96 \times 0.21$ : 95% interval [0.25, 1.08]
- ▣ Annualising by  $\sqrt{A}$  ( $A = 12$  months per year) is exact only without autocorrelation; in general  $SR(A) = \eta(A) SR$

$$\eta(A) = \frac{A}{\sqrt{A + 2 \sum_{k=1}^{A-1} (A - k) \rho_k}}$$

- ▶  $\rho_k$ : autocorrelation of monthly returns at lag  $k$ ; positive  $\rho_k$  lower  $\eta$ , negative  $\rho_k$  raise it;  $\rho_k = 0$  gives  $\eta = \sqrt{A}$
- ▣ Here  $\rho_1 = -0.03$ ,  $\sum_{k=1}^{11} \rho_k = -0.11$ :  $\eta = 3.62$  instead of  $\sqrt{12} = 3.46$ , annual Sharpe 0.69

## Testing a Difference in Sharpe Ratios

- $H_0: SR_1 = SR_2$  for two strategies evaluated on the same months; [Jobson & Korkie \(1981\)](#): a test under i.i.d. Normal returns
- [Ledoit & Wolf \(2008\)](#): a robust version
  - ▶  $r_{1t}, r_{2t}$ : excess returns of the two strategies;  $\mu_1, \mu_2$ : their means;  $\hat{\Delta} = \widehat{SR}_1 - \widehat{SR}_2 = f(\hat{v})$ ,  
 $v = (\mu_1, \mu_2, \mathbb{E}[r_1^2], \mathbb{E}[r_2^2])$ ,  $f = \frac{\mu_1}{\sqrt{\mathbb{E}r_1^2 - \mu_1^2}} - \frac{\mu_2}{\sqrt{\mathbb{E}r_2^2 - \mu_2^2}}$
  - ▶ Standard error by the delta method:  $\sqrt{\nabla f^\top \hat{\Psi} \nabla f / T}$ ,  $\nabla f$  the gradient of  $f$ ,  $\hat{\Psi}$  a HAC (heteroskedasticity and autocorrelation consistent) covariance of the four moments
- Strategies on the same assets are highly correlated: small differences can be significant

## The Block Bootstrap

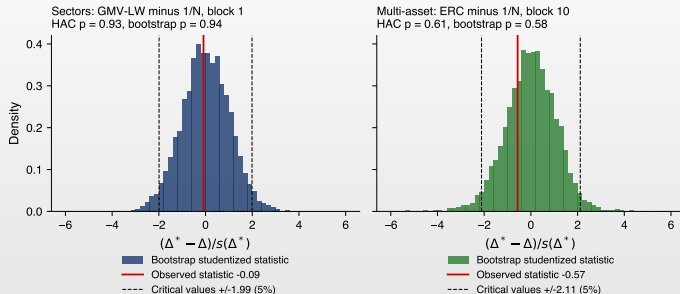


Bradley Efron, inventor of the bootstrap (1979)

Photo: Ryan K. Morris (2007); public domain; Wikimedia Commons

- Bootstrap: resample the data to approximate the sampling distribution of  $\hat{\Delta}$
- Returns are dependent (volatility clustering): resample **blocks** of consecutive months (Künsch (1989))
- Ledoit & Wolf (2008), Section 3.2.2: circular blocks of **pairs**  $(r_{1t}, r_{2t})$ , wrapping around the end of the sample, so the correlation of the two strategies is kept
- Centre and studentize: bootstrap  $|\hat{\Delta}^* - \hat{\Delta}|/s(\hat{\Delta}^*)$ ; asymptotically pivotal, more accurate than the Normal approximation of the HAC test
- $p = [\#\{\tilde{d}^{*m} \geq |\hat{\Delta}|/s(\hat{\Delta})\} + 1]/(M + 1)$ ,  $M = 4999$  draws;  $\tilde{d}^{*m}$ : the studentized statistic in draw  $m$ ,  $s(\cdot)$ : standard error,  $\#\{\cdot\}$ : number of draws
- Block length chosen by the calibration of Ledoit & Wolf (2008), Algorithm 3.1 (Appendix): mostly 1 for sectors, 8–10 for multi-asset and combined rules

## Is Anything Better than $1/N$ ?



- Bootstrap distributions of the studentized difference against  $1/N$ ; red: observed statistic; dashed: bootstrap critical values
- Sectors, GMV-LW minus  $1/N$ :  $-0.02$ , 95% interval  $[-0.36, 0.33]$ , HAC  $p = 0.93$ , bootstrap  $p = 0.94$  (block 1); multi-asset, ERC minus  $1/N$ :  $-0.03$ ,  $[-0.15, 0.08]$ , HAC  $p = 0.61$ , bootstrap  $p = 0.58$  (block 10): so correlated with  $1/N$  that the interval is narrow, yet it covers zero
- Sectors: no rule differs from  $1/N$  at 5% (closest ERC,  $+0.04$ , HAC  $p = 0.09$ ); multi-asset: MV-LO  $+0.29$  is **not** significant (HAC  $p = 0.12$ , bootstrap  $0.16$ )



## Many Rules, Many Tests

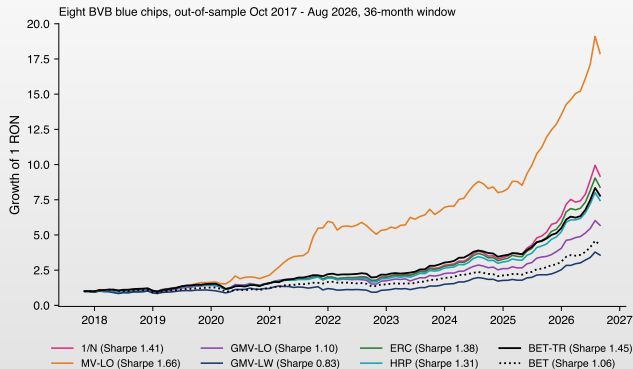
| Universe    | Rule minus 1/N | $\hat{\Delta}$ | HAC $p$ | Bootstrap $p$ |
|-------------|----------------|----------------|---------|---------------|
| Multi-asset | MV             | -0.74          | 0.01    | 0.06          |
| Multi-asset | HRP            | -0.16          | 0.07    | 0.05          |
| Combined    | GMV            | -0.57          | 0.03    | 0.05          |
| Combined    | GMV-LO         | -0.40          | 0.05    | 0.08          |
| Combined    | GMV-LW         | -0.56          | 0.02    | 0.03          |
| Combined    | MV-LO          | +0.05          | 0.82    | 0.82          |

- No rule beats 1/N significantly in any universe; unadjusted, GMV rules lose to 1/N on the combined universe: low risk, too little return
- The family:  $H_0: \Delta_k = SR_k - SR_{1/N} \leq 0$  for all  $k$ ; 21 tests (7 rules  $\times$  3 universes); Holm-adjusted  $p$ -values are at least 0.29 (HAC) and 0.68 (bootstrap): nothing survives ([Chapter 3](#), multiple testing)
- A different question, the best of 24 configurations: the deflated Sharpe ratio of [Bailey & López de Prado \(2014\)](#) gives  $0.84 < 0.95$  for MV-LO on 16 ETFs, but the verdict hinges on the effective number of trials ([Appendix](#))

## Setup: a Romanian Investor

- ▣ Eight blue chips listed since 2014: TLV, SNP, BRD, TGN, SNG, SNN, EL, TEL
- ▣ Monthly returns in RON, October 2014 – August 2026; 36-month window, first portfolio October 2017
- ▣ Rules: 1/N, MV-LO, GMV-LO, GMV-LW, ERC, HRP; benchmarks BET-TR and BET
- ▣ Sharpe ratios with  $R_f = 0$  (no RON money-market series in the data): return per unit of risk, higher than excess-return Sharpe ratios
- ▣ Earlier Romanian evidence on diversification: [Dragotă, Șerbănescu & Pele \(2008\)](#)

## BVB Portfolios versus BET-TR



- Out of sample, October 2017 – August 2026 (107 months), before costs,  $R_f = 0$
- BET-TR: mean 24.7% a year, volatility 17.0%, Sharpe 1.45; BET: mean 17.8%, Sharpe 1.07; the gap of about 7% a year is the dividends: compare with BET-TR, never with BET
- 1/N 1.41, ERC 1.38, HRP 1.31: close to BET-TR
- MV-LO 1.66 (difference  $+0.21$ ,  $p = 0.44$ ); GMV-LW 0.83 ( $-0.63$ ,  $p = 0.05$ )
- Average correlation of the eight stocks 0.44: little room for diversification inside the BVB; the latest GMV-LO puts 70% in TLV
- Open question: does diversification **beyond** BET pay? (Seminar 4, C1)

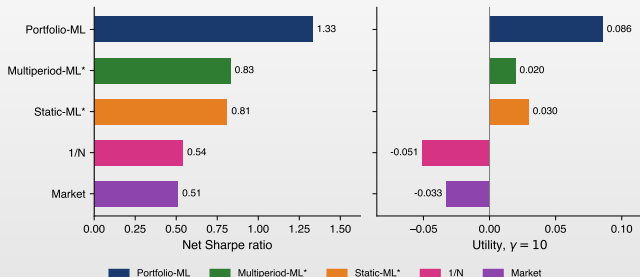
## Case Study: Which Frontier Can a Large Investor Reach?

- Jensen, Kelly, Malamud & Pedersen (2026), *Review of Financial Studies*: how should portfolios be built from machine-learning return forecasts when every trade moves prices?
- **Implementable efficient frontier** (eq. 12): for each volatility  $\sigma$ , the highest expected return **net of trading costs**,  $k(\sigma) = \max_{\pi} E[r_{t+1}^{\text{net}}]$  s.t.  $E[\pi_t^\top \Sigma \pi_t] = \sigma^2$ 
  - ▶  $\pi_t$ : vector of risky-asset weights at  $t$ ;  $r_{t+1}^{\text{net}}$ : portfolio return after trading costs;  $k(\sigma)$ : the best net expected return at volatility  $\sigma$
  - ▶ Proposition 1: the net Sharpe ratio  $k(\sigma)/\sigma$  falls along the frontier, and a larger investor faces a lower frontier
- Cost as a fraction of wealth  $W_t$  (eqs. 4, 7):  $c_t = \frac{W_t}{2}(\pi_t - g_t \pi_{t-1})^\top \Lambda (\pi_t - g_t \pi_{t-1})$ , quadratic in the trade;  $g_t$  drifts last month's weights,  $\Lambda = \text{diag}(0.2/V_i)$ ,  $V_i$  the dollar volume
- Optimal rule: trade only part of the way towards an **aim portfolio**,  $\pi_t = m g_t \pi_{t-1} + (1 - m)A_t$ ,  $m$ : the share of the old position kept,  $A_t$ : the aim portfolio (Gârleanu & Pedersen (2013)); Portfolio-ML learns  $A_t$  from the signals by a ridge regression that maximises utility net of costs

|               | Paper                                                  | Course replication                       |
|---------------|--------------------------------------------------------|------------------------------------------|
| Assets        | NYSE stocks above median size (184–805)                | 14 US stocks + 9 sector ETFs             |
| Signals       | 115 characteristics (Jensen, Kelly & Pedersen (2023))  | 5 price and volume signals               |
| Out of sample | 1981–2020, monthly                                     | February 2014 – August 2026 (151 months) |
| Investor      | $\gamma = 10$ , wealth \$10 billion at the end of 2020 | same                                     |

## Case Study: Table 2 at a Glance

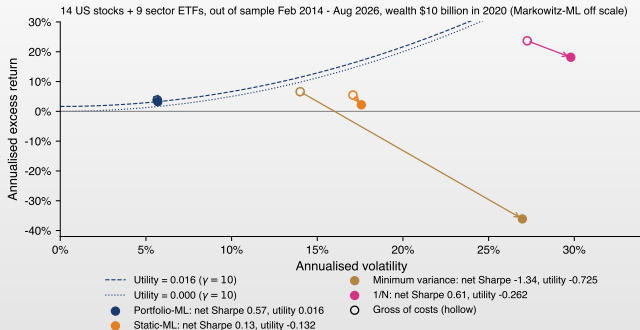
Out of sample 1981–2020, US stocks, wealth \$10 billion (Markowitz-ML off scale)



- Jensen, Kelly, Malamud & Pedersen (2026), Table 2: net of costs, 1981–2020,  $\gamma = 10$ ; utility (eq. 11) = annual net return  $-\frac{\gamma}{2} \times$  annual variance
- Markowitz-ML (mean–variance on ML return forecasts, no costs): gross Sharpe 2.06, turnover 79.36, leverage 74.61: net return extremely negative
- Portfolio-ML: net Sharpe 1.33, utility 0.086 (net return 15%, volatility 11%, cost 0.6% a year)
- Static-ML\* (one-period costs; \* = extra tuning layer) 0.81, 0.030; 1/N 0.54,  $-0.051$ ; market 0.51,  $-0.033$ : only cost-aware rules have positive utility

 MFM\_ch4\_implementable\_frontier

## Case Study: Gross and Net of Costs on the Course Data



- Hollow marker: before costs; filled: after the costs of eq. (35); dashed curve: pairs with the utility of Portfolio-ML (0.016), dotted: utility 0
- Portfolio-ML barely moves: 3.8% gross, 3.3% net, 5.7% volatility; net Sharpe 0.57
- 1/N keeps 18.1% net (Sharpe 0.61), but at 29.8% volatility its utility is  $-0.26$
- Minimum variance: +6.6% gross,  $-36.1\%$  net, rebalancing a leverage of 2.6 costs 43% a year
- Limits: 23 assets, 5 signals, no factor model for  $\Sigma$ ; the frontier that matters is the one after costs (Seminar 4, C3)

 MFM\_ch4\_implementable\_frontier

## Open Questions

- Questions
  - ▶ does any **estimated** allocation rule beat  $1/N$  out of sample, net of costs?
  - ▶ does it still beat  $1/N$  once every rule, window and universe tried is counted?
- Live evidence points both ways
  - ▶ Nonlinear shrinkage of  $\Sigma$  improves GMV out of sample in large cross-sections: [Ledoit & Wolf \(2017\)](#)
  - ▶ Combining tangency, GMV and  $1/N$  under estimation risk: [Kan, Wang & Zhou \(2022\)](#)
  - ▶ Backtests need haircuts for the number of specifications tried: [Harvey & Liu \(2015\)](#)
- Formal statement for  $K$  rules on one universe:  $H_0: \Delta_k = SR_k - SR_{1/N} \leq 0$  for all  $k$ ;  $H_1: \Delta_k > 0$  for some  $k$ ; error rate fixed for the whole family
- One pass of the loop with an AI assistant: literature with DOIs, competing hypotheses, code on the course's monthly data, a pre-registered family of tests, a referee-style critique (prompts in the [Appendix](#))
- Falsified by: a positive  $\hat{\Delta}_k$  that survives a family-wise correction, costs and a second sample

## What the Researcher Must Check

- **Look-ahead:** weights at  $t$  built with month- $t$  returns; assert that the last estimation date is before  $t$
- **Survivorship and selection:** an ETF list chosen today; use the universe available at each date, including later delisted ETFs, or disclose the limitation
- **Multiple testing:** 7 rules  $\times$  3 universes already give 21 tests; Holm (FWER) or Benjamini–Hochberg (FDR, [Benjamini & Hochberg \(1995\)](#)) adjusted  $p$ -values
- **Power:**  $\hat{\Delta}$  has a standard error near 0.18 on 171 months; compute the detectable effect before running
- **Return type:** portfolios aggregate **simple** returns, Sharpe ratios use **excess** returns; recompute one month step by step
- **Invented references:** an unresolvable DOI or a title mismatch drops the claim and goes into `AI_ERRORS.md`



## Mini-Case: Data Snooping in This Chapter's Backtest

- ▣ A typical assistant proposal: “MV-LO beats 1/N on the multi-asset universe”; superiority is one-sided:  $H_0: \Delta \leq 0$ ,  $p = 1 - \Phi(\hat{\Delta}/\text{se})$ ,  $\Phi$  the standard Normal distribution function, se the standard error
- ▣ Replicate:  $\hat{\Delta} = +0.29$ , HAC standard error 0.18, one-sided  $p = 0.06$  (two-sided HAC 0.12, bootstrap 0.16)
- ▣ Test the family, not the winner: 21 tests (7 rules  $\times$  3 universes)
  - ▶ Only 7 of 21 differences are positive; the 4 two-sided HAC  $p$ -values below 0.05 are all **losses**, which never reject a superiority null
  - ▶ One-sided: smallest  $p = 0.046$  (sectors, ERC); adjusted Holm 0.96, Benjamini–Hochberg 0.62; 0 discoveries at an FDR of 10%
- ▣ Power 0.8 for  $\Delta = 0.29$ , one-sided: about 435 months (36 years) at 5%, 944 months (79 years) at the Holm level 5%/21
- ▣ Interpretation: the data cannot rank these rules; the bootstrap interval for MV-LO,  $[-0.13, 0.70]$ , admits losses and large gains; failure to reject is not proof of equality

## Your Turn: A Research Project Idea

### ▣ Questions

- ▶ does nonlinear shrinkage (Ledoit & Wolf (2017)) beat  $1/N$  on the 16 ETFs of this chapter, after 10–50 bp costs?
- ▶ does a three-fund combination (Kan, Wang & Zhou (2022)) beat  $1/N$  on the same ETFs, after the same costs?

- ▣ **Replicate first:** the combined-universe results ( $1/N$  Sharpe 0.76, MV-LO 0.81) and the adjusted  $p$ -values of the mini-case

- ▣ **Pre-register:** hypotheses, family of tests, error rate, power and a placebo (random long-only weights with the same turnover), fixed before any result

- ▣ **Report:** code on the course's monthly data, a table of all specifications, a referee-style self-critique

- ▣ **Log:** prompts in AI\_USE.md; every error caught and the test that caught it in AI\_ERRORS.md

## Conclusions: Which Rule, When?

| Rule            | Objective                    | Inputs                         | Main failure mode                     | Evidence in this chapter                          |
|-----------------|------------------------------|--------------------------------|---------------------------------------|---------------------------------------------------|
| MV              | maximum Sharpe               | $\hat{\mu}, \hat{\Sigma}^{-1}$ | error maximisation, explosive weights | worst in every universe; insolvent path           |
| MV-LO           | maximum Sharpe, $w_i \geq 0$ | $\hat{\mu}, \hat{\Sigma}$      | corner solutions, turnover            | best point estimates, never significant           |
| GMV, GMV-LW     | minimum variance             | $\hat{\Sigma}$ (shrunk)        | loads on low-return assets            | lowest risk, loses to 1/N on 16 ETFs (unadjusted) |
| ERC, HRP        | balanced risk                | $\hat{\Sigma}$                 | a tilt of 1/N; HRP trades more        | never significantly better than 1/N               |
| Black–Litterman | benchmark plus views         | $w_b$ , views, $\Omega$        | confidence set too high               | moves only the assets in the views                |
| 1/N             | none                         | none                           | ignores risk differences              | hard to beat after costs                          |

- Evaluate every rule out of sample, net of costs, with a test of the Sharpe difference and a count of all tests; these verdicts are specific to these samples

## Check Yourself, and Next

### Retrieval questions

- Why does the in-sample tangency Sharpe overstate the achievable one?
- By how much does it overstate it for  $N = 9$ ,  $T = 60$ ?
- Why does a long-only constraint act like shrinkage?
- Why can a small Sharpe difference between ERC and  $1/N$  be nearly significant while a larger one between GMV and  $1/N$  is not?

### Links to other chapters

- **Chapter 3**: the tangency portfolio and the CAPM; factor covariance models
- **Chapters 5–6**: time-varying volatility and correlation for dynamic weights
- **Chapter 7**: portfolio Value-at-Risk and Expected Shortfall
- **Chapter 8**: backtesting without overfitting
- **Chapter 13**: machine learning for returns and portfolios

## Appendix: Wishart Moments and the Distribution of Weights

- i.i.d.  $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  returns, maximum-likelihood estimators:  $\hat{\boldsymbol{\mu}} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma}/T)$ ,  $T\hat{\boldsymbol{\Sigma}} \sim W_N(T-1, \boldsymbol{\Sigma})$  (Wishart: the matrix analogue of  $\chi^2$ ,  $T-1$  degrees of freedom), independent
- Inverse Wishart moment, finite for  $T > N+2$ :  $\mathbb{E}[\hat{\boldsymbol{\Sigma}}^{-1}] = \frac{T}{T-N-2}\boldsymbol{\Sigma}^{-1}$ , an **upward** bias that grows with  $N/T$
- Conditioning on  $\hat{\boldsymbol{\mu}}$  and using  $\mathbb{E}[\mathbf{x}^\top \mathbf{K} \mathbf{x}] = \text{tr}(\mathbf{K} \text{Var}(\mathbf{x})) + \mathbb{E}[\mathbf{x}]^\top \mathbf{K} \mathbb{E}[\mathbf{x}]$  (any random vector  $\mathbf{x}$ , fixed matrix  $\mathbf{K}$ ,  $\text{tr}$ : trace) gives  $\mathbb{E}[\hat{\theta}^2]$  of the lecture (Seminar A9)
- GMV weights with Wishart  $\hat{\boldsymbol{\Sigma}}$  and  $T$  large enough relative to  $N$ :  $\hat{\mathbf{w}}_{\text{GMV}}$  is unbiased and its exact distribution and covariance are known (Okhrin & Schmid (2006))
- Tangency weights divide by  $1^\top \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\mu}}$ , which can be near zero: very heavy tails
- Estimation-risk-optimal mixtures of tangency, GMV and  $1/N$ : Kan & Zhou (2007), Kan, Wang & Zhou (2022)

## Appendix: Tangency Weights by Regression, Britten-Jones (1999)

- Britten-Jones (1999): regress the constant 1 on the excess returns, **without** a constant term

$$1 = \mathbf{b}^\top \mathbf{r}_t + u_t, \quad \hat{\mathbf{b}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{1}_T \propto \hat{\Sigma}^{-1} \hat{\boldsymbol{\mu}}$$

- $\mathbf{r}_t$ : vector of excess returns in month  $t$ ;  $\mathbf{X}$ :  $T \times N$  matrix with rows  $\mathbf{r}_t^\top$ ;  $\mathbf{1}_T$ : vector of  $T$  ones;  $u_t$ : regression error
- Sherman–Morrison on  $\mathbf{X}^\top \mathbf{X} / T = \hat{\Sigma} + \hat{\boldsymbol{\mu}} \hat{\boldsymbol{\mu}}^\top$  gives the proportionality (Seminar A2);  $\hat{\mathbf{b}} / \mathbf{1}^\top \hat{\mathbf{b}} =$  sample tangency weights if  $\mathbf{1}^\top \hat{\mathbf{b}} > 0$
- OLS  $t$ -test of  $b_i = 0$ : asset  $i$  has zero weight in the tangency portfolio;  $F$ -test of  $\mathbf{R}\mathbf{b} = 0$  with  $\mathbf{R} = [\mathbf{I}_{N-1}, -\mathbf{1}_{N-1}]$ : “the tangency portfolio is  $1/N$ ”
- Exact finite-sample  $t$  and  $F$  distributions under i.i.d. Normal returns; HAC standard errors otherwise
- Nine sectors, January 1999 – July 2026 ( $T = 331$ ): normalised  $\hat{\mathbf{b}}$  reproduces the tangency weights; largest  $|t| = 1.12$  (financials), HAC 1.19
- $H_0$ : tangency =  $1/N$ , eight restrictions:  $F(8, 322) = 0.48$ ,  $p = 0.87$ ; HAC Wald  $\chi^2_8 = 4.50$ ,  $p = 0.81$ ; not rejecting is not proof of equal weights

## Appendix: The Optimal Shrinkage Intensity

- Ledoit–Wolf choose  $\delta$  to minimise the expected squared (Frobenius) distance to the true  $\Sigma$

$$\delta^* = \max \left\{ 0, \min \left\{ 1, \frac{\hat{\kappa}}{T} \right\} \right\}, \quad \hat{\kappa} = \frac{\hat{\pi} - \hat{\rho}}{\hat{\gamma}}$$

- $\hat{\pi}$ : sum of the asymptotic variances of the entries of  $S$  (noise)
- $\hat{\rho}$ : their asymptotic covariances with the target entries
- $\hat{\gamma} = \|F - S\|^2$ : misspecification of the target
- Complete formulas for the constant-correlation target: [Ledoit & Wolf \(2004b\)](#), Appendix; code: the [shrinkage Quantlet](#)

## Appendix: Shrinking the Means and Other Covariance Estimators

- Bayes–Stein ([Jorion \(1986\)](#)):  $\hat{\mu}_{BS} = (1 - \hat{\phi}) \hat{\mu} + \hat{\phi} \hat{\mu}_{GMV}$ 
  - ▶ Empirical-Bayes shrinkage of every mean towards the mean of the GMV portfolio;  $\hat{\phi}$  from a Stein argument, larger when  $N/T$  is large
- [Tu & Zhou \(2011\)](#): the convex combination of a sophisticated rule and  $1/N$  that minimises the expected out-of-sample loss
- [DeMiguel, Garlappi, Nogales & Uppal \(2009\)](#): norm constraints  $\|w\|_1 \leq \delta$  or  $\|w\|_2 \leq \delta$  nest the long-only GMV, shrinkage GMV and  $1/N$ ; the bridge to the lasso portfolios of [Chapter 13](#)
- Factor structure plus a sparse residual covariance: POET ([Fan, Liao & Mincheva \(2013\)](#)), the link to [Chapter 3](#)
- High-dimensional GMV with shrinkage towards a target portfolio: [Bodnar, Parolya & Schmid \(2018\)](#)



## Appendix: Black–Litterman as Mixed (GLS) Estimation

- ▣ Theil & Goldberger (1961): stack the prior and the views as one regression on  $\mu$

$$\begin{pmatrix} \pi \\ q \end{pmatrix} = \begin{pmatrix} I \\ P \end{pmatrix} \mu + \begin{pmatrix} e_1 \\ e_2 \end{pmatrix}, \quad \text{Var} \begin{pmatrix} e_1 \\ e_2 \end{pmatrix} = \begin{pmatrix} \tau \Sigma & 0 \\ 0 & \Omega \end{pmatrix}$$

- ▣ GLS gives exactly  $\mu_{BL}$ , with covariance  $M = [(\tau \Sigma)^{-1} + P^\top \Omega^{-1} P]^{-1}$
- ▣ Equivalent form:  $\mu_{BL} = \pi + \tau \Sigma P^\top (P \tau \Sigma P^\top + \Omega)^{-1} (q - P \pi)$
- ▣  $\Omega \rightarrow 0$ : restricted least squares,  $P \mu_{BL} = q$
- ▣ Predictive covariance  $\Sigma + M$  in the weights would change all weights; without views it scales them by  $1/(1 + \tau)$

## Appendix: Other Risk-Based Rules

- ▣ **Inverse volatility:**  $w_i \propto 1/\sigma_i$ ; equals ERC when all correlations are equal
- ▣ **Maximum diversification (Choueifaty & Coignard (2008)):** maximise  $w^\top \sigma / \sqrt{w^\top \Sigma w}$ , the ratio of the average asset volatility to the portfolio volatility;  $\sigma$ : vector of asset volatilities
- ▣ **Minimum variance:** GMV, the extreme risk-based rule
- ▣ All ignore expected returns: robust to estimation error, but optimal only under special assumptions

## Appendix: Results, Sector Universe

| Rule   | Mean  | Volatility | Sharpe | Turnover | Max. drawdown |
|--------|-------|------------|--------|----------|---------------|
| 1/N    | 9.4%  | 14.3%      | 0.66   | 0.02     | −49%          |
| MV     | 17.8% | 69.0%      | 0.26   | 3.86     | −100%         |
| MV-LO  | 10.7% | 14.7%      | 0.72   | 0.23     | −42%          |
| GMV    | 8.5%  | 11.9%      | 0.71   | 0.23     | −33%          |
| GMV-LO | 8.1%  | 11.5%      | 0.70   | 0.09     | −34%          |
| GMV-LW | 7.6%  | 11.8%      | 0.65   | 0.14     | −31%          |
| ERC    | 9.3%  | 13.2%      | 0.70   | 0.03     | −47%          |
| HRP    | 8.9%  | 12.8%      | 0.70   | 0.08     | −45%          |

- ▣ January 2004 – July 2026; turnover: mean monthly traded fraction of wealth; drawdown of total-return wealth
- ▣ MV: a monthly loss above 100% wipes out the capital (drawdown −100%); its mean and Sharpe describe a hypothetical return series

## Appendix: Results, Combined Universe (16 ETFs)

| Rule   | Mean  | Volatility | Sharpe | Turnover | Sharpe, 50 bp |
|--------|-------|------------|--------|----------|---------------|
| 1/N    | 7.9%  | 10.3%      | 0.76   | 0.02     | 0.75          |
| MV     | 28.7% | 261.8%     | 0.11   | 22.10    | -0.51         |
| MV-LO  | 8.0%  | 9.8%       | 0.81   | 0.20     | 0.69          |
| GMV    | 0.8%  | 4.1%       | 0.20   | 0.24     | -0.15         |
| GMV-LO | 2.0%  | 5.4%       | 0.36   | 0.06     | 0.30          |
| GMV-LW | 0.9%  | 4.4%       | 0.20   | 0.13     | 0.03          |
| ERC    | 5.5%  | 7.8%       | 0.71   | 0.03     | 0.68          |
| HRP    | 4.0%  | 7.0%       | 0.56   | 0.11     | 0.47          |

▣ May 2012 – July 2026; last column: net of 50 basis points per unit of turnover

## Appendix: Bootstrap Details and the Block Length

- Studentization:  $s(\hat{\Delta}^*)$  from the block sums of each resample;  $s(\hat{\Delta})$  on the data from a prewhitened quadratic-spectral kernel ([Andrews & Monahan \(1992\)](#)) with automatic bandwidth ([Andrews \(1991\)](#))
- [Ledoit & Wolf \(2008\)](#), Algorithm 3.1: choose the block  $b$  whose bootstrap interval has actual coverage closest to 95%
  - ▶ Fit a first-order vector autoregression to the pair of returns; resample its residuals by a stationary bootstrap ([Politis & Romano \(1994\)](#)), mean block 5
  - ▶ Generate  $K = 5000$  pseudo-samples of length  $T$ ; in each, build the 95% interval for  $b \in \{1, 2, 4, 6, 8, 10\}$
  - ▶  $\hat{g}(b)$ : share of intervals that contain the pseudo-true  $\hat{\Delta}$ ; pick  $\tilde{b} = \arg \min_b |\hat{g}(b) - 0.95|$
- The grid and  $K$  are those of the paper (Sections 4–5); applied to each of the 21 pairs of this chapter
- A non-studentized bootstrap with a fixed 6-month block is not this test: its block is not calibrated for coverage

## Appendix: The Deflated Sharpe Ratio

- Bailey & López de Prado (2014): the best of  $N$  Sharpe ratios is biased upwards even if every true Sharpe ratio is zero
- Expected maximum under  $H_0$ : all true Sharpe ratios zero,  $N$  independent trials, monthly units as  $T$ ; with  $\gamma_E = 0.5772$  (Euler's constant) and  $\hat{V}$  the variance of the  $N$  estimates;  $\Phi$ : standard Normal distribution function,  $e$ : Euler's number

$$SR_0 = \sqrt{\hat{V}} \left[ (1 - \gamma_E) \Phi^{-1} \left( 1 - \frac{1}{N} \right) + \gamma_E \Phi^{-1} \left( 1 - \frac{1}{Ne} \right) \right]$$

- Deflated Sharpe ratio: the probability that the true Sharpe of the winner exceeds  $SR_0$ , with skewness  $\hat{\gamma}_3$  and kurtosis  $\hat{\gamma}_4$  of the winner

$$DSR = \Phi \left( \frac{(\widehat{SR} - SR_0) \sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}^2}} \right)$$

- This chapter: 24 configurations (8 rules  $\times$  3 universes); winner MV-LO on 16 ETFs, 0.81 a year
  - ▶ As 24 independent trials:  $SR_0 = 0.52$  a year,  $DSR = 0.84 < 0.95$
  - ▶ The configurations are strongly correlated (up to 0.99), so fewer are independent:  $N = 8$  gives DSR 0.93,  $N = 3$  gives 0.98
- Holm adjusts the whole family of pairwise tests against  $1/N$ ; the DSR asks whether the single best configuration is better than luck

## Appendix: Beyond Equities, Digital Assets

- ▣ [Petukhina, Trimborn, Härdle & Elendner \(2021\)](#): cryptocurrencies in portfolio allocation rules; gains depend on the rule and on liquidity
- ▣ [Pele et al. \(2023\)](#): are cryptos becoming alternative assets?
- ▣ Bitcoin: annualised volatility 172% (monthly, 2010–2026, one surviving coin chosen ex post), IEF 6.6%
- ▣ Risk contributions grow with volatility: risk-based rules give Bitcoin a small weight automatically (Seminar 4, B5)
- ▣ Robust mean–variance for crypto-assets: [Pele \(2023\)](#)

## Appendix: Case Study, Method in Detail (1/2)

- Cost (eqs. 4, 7):  $\pi_t$  risky-asset weights,  $1 - \mathbf{1}^\top \pi_t$  in the risk-free asset;  $g_t = \text{diag}((1 + r_t^f + r_t)/(1 + g_t^w))$  (eq. 6),  $g_t^w$  the growth of wealth;  $\Lambda = \text{diag}(0.2/V_i)$ ,  $V_i$  the average daily dollar volume over 6 months: trading 1% of  $V_i$  moves the price by 0.1% (eq. 35, [Frazzini, Israel & Moskowitz \(2018\)](#))
- Optimal rule (Proposition 3):  $\pi_t = m g_t \pi_{t-1} + (1 - m) A_t$ ,  $A_t$  a weighted sum of today's and expected future Markowitz portfolios;  $m$  solves eq. (14); [Gârleanu & Pedersen \(2013\)](#) generalised
- Portfolio-ML (Proposition 4, eq. 40):  $A_t = \text{diag}(1/\sigma_{i,t}) f(s_t) \beta$ ,  $f = p$  random Fourier features of the signals  $s_t$ ;  $\hat{\beta} = (E_T[\tilde{\Sigma}_t] + \lambda I)^{-1} E_T[\tilde{r}_{t+1}]$ , a ridge regression that maximises utility net of costs in closed form (eqs. 24–26), learning the weights directly as [Brandt, Santa-Clara & Valkanov \(2009\)](#) do without costs



## Appendix: Case Study, Method in Detail (2/2)

- Data (Section 5.1): NYSE common stocks above the median market capitalisation (184–805 stocks), 115 characteristics of Jensen, Kelly & Pedersen (2023) ranked into  $[0, 1]$ ; December 1952 – December 2020, out of sample 1981–2020, monthly
- $\Sigma$ : 13 characteristic themes and 12 industries as factors, exponential weights with half-lives of 378 days (correlations) and 126 days (variances); investor  $\gamma = 10$ , wealth \$10 billion at the end of 2020, growing with the market (Section 5.1.3)
- Tuning (Table 1):  $\lambda \in \{0, e^{-10}, e^{-9.8}, \dots, e^{10}\}$ ,  $p \in \{2^6, \dots, 2^9\}$ ,  $\eta \in \{e^{-3}, e^{-2}\}$  (standard deviation of the random weights), chosen each year by the utility of validation backtests since 1971
- Rivals: Markowitz-ML (eq. 32), Static-ML (one-period cost, eq. 34), Multiperiod-ML (forecasts for 12 horizons, eq. 22), factor sorts, 1/N, minimum variance, market; “\*” marks an extra tuning layer (Section 5.3)

## Appendix: Case Study, Complete Results

- Table 2, Portfolio-ML: net Sharpe ratio 1.33, utility 0.086 (net return 15%, volatility 11%, cost 0.6% a year, turnover 0.25)
- Table 2, Multiperiod-ML\*: net Sharpe 0.83, utility 0.020; Portfolio-ML needs no second tuning layer
- Table 3: posterior probability that Portfolio-ML has the higher utility: 99.9% against Static-ML\* and Multiperiod-ML\*, 100% against Markowitz-ML
- Figure 1(B) (Section 6.2): with \$10 billion and  $\gamma = 10$ , net excess return 14.6% at 11% volatility; with \$1 billion, 16.3% at the same volatility: size is a cost
- Figure 5: net of costs, quality is the most important theme and short-term reversal almost irrelevant; monthly autocorrelation  $-0.04$  for 1-month reversal,  $0.94$  for book-to-market (Section 6.3): persistent signals are cheap to trade
- Limits stated by the authors: fees and other frictions, computing power and ML methods not available in real time early in the sample (Section 6.1); costs and risk treated as constant over the horizon of each optimisation (Section 5.1.3); among the smallest 20% of stocks Portfolio-ML ranks behind Multiperiod-ML\* and Static-ML\* (Figure 8)

## Appendix: Case Study, the Course Replication

- Course data: 14 US stocks and 9 sector ETFs; five signals (1-month return, momentum from  $t - 6$  and from  $t - 12$  to  $t - 1$ , 1-year volatility, 6-month dollar volume); same cost rule,  $\gamma$ , wealth and grid;  $\Sigma$  with the same weights on the 23 assets; validation from 2011; out-of-sample returns February 2014 – August 2026 (151 months)
- Portfolio-ML: net Sharpe 0.57, utility 0.016, cost 0.6% a year, leverage 0.32; 1/N: net Sharpe 0.61, utility  $-0.26$  (net volatility 30%)
- Utility: Static-ML  $-0.13$ ; minimum variance  $-0.72$  (cost 43% a year); Markowitz-ML extremely negative (leverage 18)
- Posterior probability of a higher utility for Portfolio-ML: 99.5% against Static-ML, 99.9% against minimum variance, 93.3% against 1/N
- Same ranking of cost-aware against cost-agnostic rules as Table 2; Portfolio-ML wins with small, slowly traded positions, not with a higher gross Sharpe ratio (0.68 against 0.87 for 1/N)
- Limits: 23 assets instead of 184–805, five signals instead of 115, no factor model for  $\Sigma$ , whose window reaches 10 years only in August 2014 (common prices from August 2004); return forecasts with  $p \geq 2^6$  and three validation years (paper:  $2^1$ – $2^9$ , ten years), no starred rivals
- Open extension: implementable frontiers for wealth of \$1, 10 and 100 billion and  $\gamma \in \{1, 5, 10, 20, 100\}$ , with 1/N's trading costs computed at each wealth level (Seminar 4, C3)

## Appendix: The Discovery Loop, Reusable Prompts (1/2)

- Tools are interchangeable: an AI assistant (Claude, ChatGPT, Gemini, GitHub Copilot) and literature engines (Semantic Scholar, Elicit, Google Scholar)
- **Literature:** *"List peer-reviewed studies since 2015 comparing shrinkage or risk-based rules with  $1/N$  out of sample; for each give DOI,  $N$ ,  $T$ , cost assumption and test."* Resolve every DOI on Crossref and open the table cited
- **Hypotheses:** *"Give three competing explanations for a Sharpe gain over  $1/N$  (better  $\Sigma$ , a low-volatility premium, luck) and the testable implication of each for turnover, subperiods and factor loadings."*
- **Data and code:** *"Write Python that reads the course's daily price files, aligns prices on common dates and runs a 60-month rolling GMV-LW and  $1/N$ ; weights at  $t$  may use data up to  $t - 1$  only."* Replicate first:  $1/N$  0.76, GMV-LW 0.20 on the combined universe

## Appendix: The Discovery Loop, Reusable Prompts (2/2)

- **Robustness:** *"Draft a pre-registration: the family of tests (rules  $\times$  universes  $\times$  windows  $\times$  costs), the error rate (FWER or FDR), the subperiods and a placebo, fixed before any result is seen."*
- **Critique:** *"Act as a referee: give the three strongest reasons my Sharpe gain is spurious and the test that would expose each."*
- **Report:** every specification tried, adjusted  $p$ -values, the power of the design, code and seeds; any specification chosen after seeing results is exploratory
- Portfolios aggregate **simple** returns and Sharpe ratios use **excess** returns: recompute one month step by step

## Appendix: Further Study, Code and Data

### Advanced courses on quantinar.com

- ☐  Hierarchical Clustering
- ☐  Minimum Spanning Tree
- ☐  Multivariate Statistical Analysis

### Code and data

- ☐ All charts and tables: [github.com/danpele/MFM/tree/main/Quantlets/Ch\\_04](https://github.com/danpele/MFM/tree/main/Quantlets/Ch_04)
- ☐ Risk-free rate: [Kenneth French Data Library](#)

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